Ethan Krul

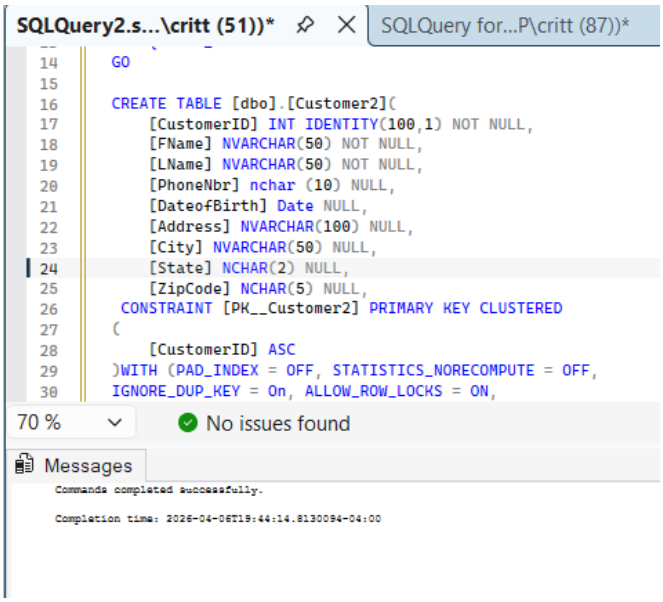
ISIN 325

4/6/2026

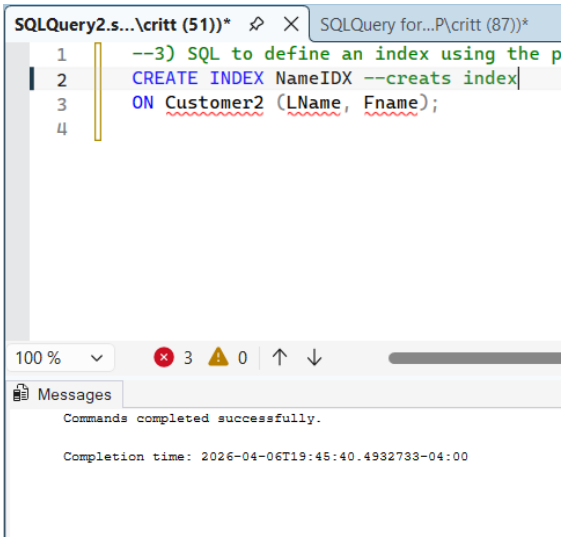
SQL3

1) SQL to create a Customer2 table using appropriate data types that has the following columns: CustomerID, First Name, Last Name, Date of Birth, Address, City, State, Zip Code. CustomerID must be defined as an identity field that starts with 100.

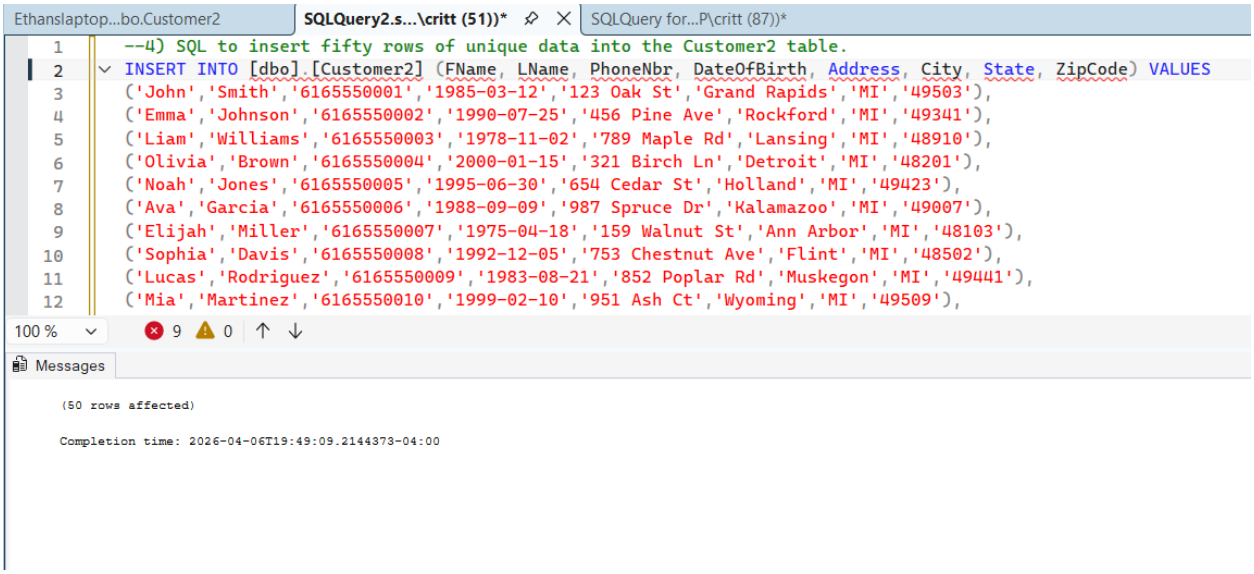
2) SQL to define the primary key in the CustomerID column. (SQL must not be generated)



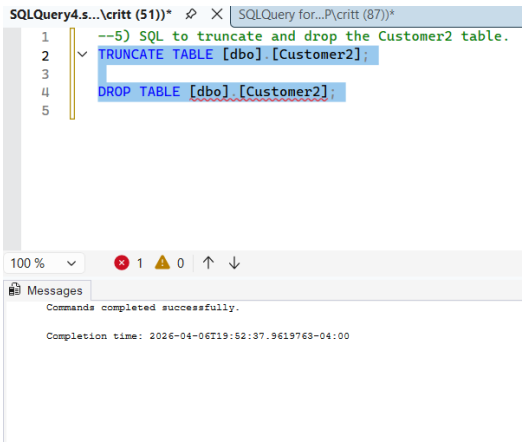
3) SQL to define an index using the primary key and a second index using Last Name and First Name. (SQL must not be generated)



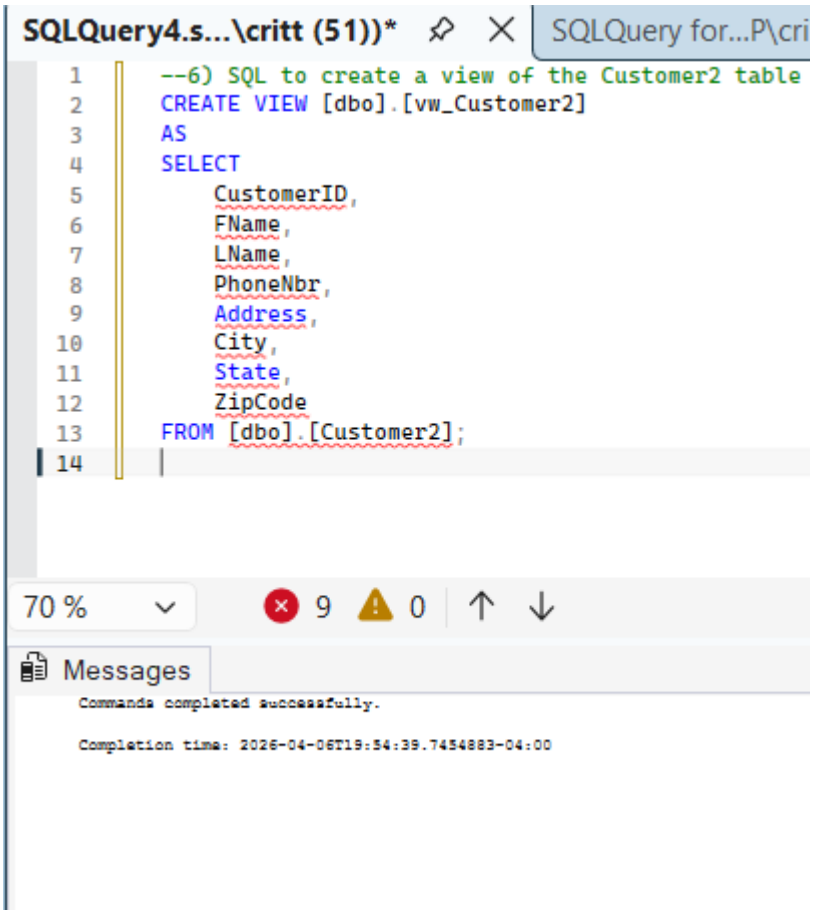
4) SQL to insert fifty rows of unique data into the Customer2 table.



5) SQL to truncate and drop the Customer2 table.



6) SQL to create a view of the Customer2 table using all of the columns in the table except Date of Birth.



7) SQL that uses the view to display all Customers in the Customer2 table sorted by First Name in descending order.

SQLQuery4.s...\critt (51))* | SQLQuery for...P\critt (87))*

```
1  --7) SQL that uses the view to display all Customers in the Customer2 table sorted by Firs
2  SELECT *
3  FROM [dbo].[vw_Customer2]
4  ORDER BY FName DESC;
```

70 % 2 0

Results | Messages

	CustomerID	FName	LName	PhoneNbr	Address	City	State	ZipCode
1	139	Zoey	Nelson	6165550040	676 Berry St	Eastpointe	MI	48021
2	131	Victoria	Young	6165550032	707 Pineapple St	Shelby Twp	MI	48315
3	107	Sophia	Davis	6165550008	753 Chestnut Ave	Flint	MI	48502
4	128	Sebastian	Lewis	6165550029	404 Melon St	Warren	MI	48089
5	125	Scarlett	Sanchez	6165550026	101 Berry Ln	Southfield	MI	48075
6	138	Samuel	Adams	6165550039	565 Peach Dr	Roseville	MI	48066
7	147	Riley	Evans	6165550048	555 Orange Rd	Taylor	MI	48180
8	103	Olivia	Brown	6165550004	321 Birch Ln	Detroit	MI	48201
9	141	Nora	Campbell	6165550042	898 Plum Rd	Grosse Pointe	MI	48230
10	104	Noah	Jones	6165550005	654 Cedar St	Holland	MI	49423

Now use the Meijer database ->

8) One sql routine with at least 3 If statements and confirmation that they work.

SQLQuery4.s...\critt (51))* | SQLQuery for...P\critt (87))*

```
1  /*Now use the Meijer database -> */
2
3  --8) One sql routine with at least 3 If statements and confirmation that they work.
4  Declare @Counter int = 3
5  if @Counter < 2
6      Print 'Less than 2'
7  If @Counter = 2
8      print 'Equal 2'
9  If @Counter > 2
10     Print 'Greater than 2' + CAST(@Counter As Varchar(10))
11
```

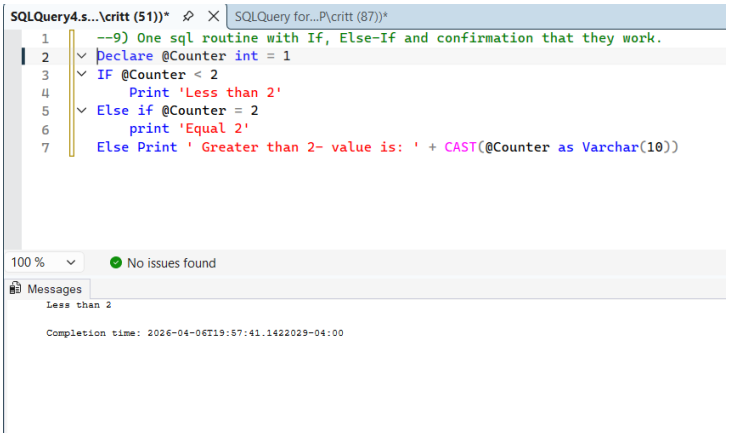
70 % No issues found

Messages

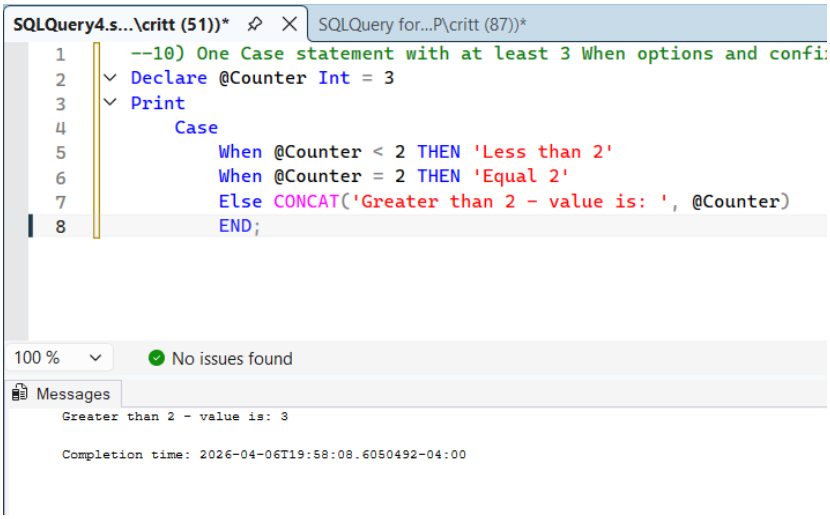
```
Greater than 23

Completion time: 2026-04-06T19:56:28.3173829-04:00
```

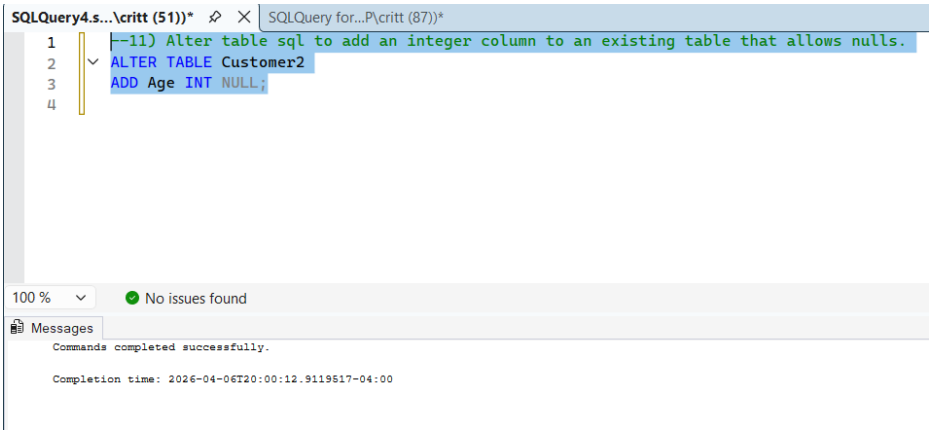
9) One sql routine with If, Else-If and confirmation that they work.



10) One Case statement with at least 3 When options and confirmation that they work.



11) Alter table sql to add an integer column to an existing table that allows nulls.



12) SQL update routine with error checking and error logging. Confirmation statement that generates logged error.

```
--12) SQL update routine with error checking and error logging.  Confirmation statement that generates logged error.
DECLARE @ErrorHold INT;
INSERT INTO Customer2 (CustomerID, FName) -- This will generate an error because CustomerID is an identity column
VALUES (51, 'John');
SET @ErrorHold = @@ERROR; -- Capture the error
PRINT CONCAT('@@error is: ', @ErrorHold); -- Print confirmation
IF @ErrorHold != 0 -- Log the error if it exists
BEGIN
    INSERT INTO ErrorAudit (ErrorNbr, ErrorTime)
    VALUES (@ErrorHold, GETDATE());
END
```

```
@@error is: 544

(1 row affected)

Completion time: 2026-04-06T20:03:11.2288660-04:00
```

13) SQL statement that defines and uses variables.

```
--13) SQL statement that defines and uses variables.
DECLARE @FirstName NVARCHAR(50); -- Define variables and use them
DECLARE @LastName NVARCHAR(50);
DECLARE @City NVARCHAR(50);
    SET @FirstName = 'Alice'; -- Assign values to the variables
    SET @LastName = 'Walker';
    SET @City = 'Grand Rapids';
        PRINT CONCAT(@FirstName, ' ', @LastName, ' lives in ', @City);-- Use the variables in a SELECT or PRINT statement
```

```
Alice Walker lives in Grand Rapids

Completion time: 2026-04-06T20:05:32.0555598-04:00
```

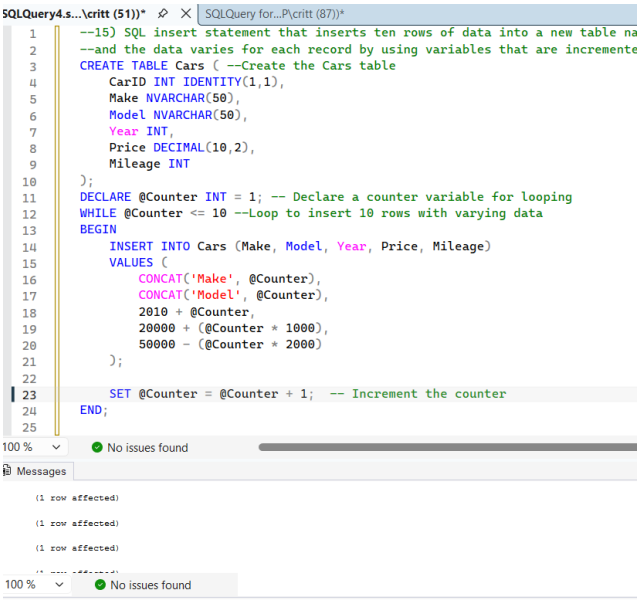
14) Store procedure that accepts a variable and has error checking and logging as part of the execute stored procedure statement.

```
--14) Store procedure that accepts a variable and has error checking and logging as part of the execute stored procedure statement.
CREATE PROCEDURE InsertCustomer -- Create the procedure
    @FName NVARCHAR(50), -- Input parameter: First Name
    @LName NVARCHAR(50) -- Input parameter: Last Name
AS
BEGIN
    DECLARE @ErrorHold INT; -- Declare a variable to hold any error codes
    INSERT INTO Customer2 (FName, LName)     -- Attempt to insert the new customer into Customer2
    VALUES (@FName, @LName);
    SET @ErrorHold = @@ERROR;     -- Capture any error that occurred during the insert
    IF @ErrorHold != 0     -- If there was an error, log it into the ErrorAudit table
    BEGIN
        INSERT INTO ErrorAudit (ErrorNbr, ErrorTime)
        VALUES (@ErrorHold, GETDATE());
    END
END;
EXEC InsertCustomer 'John', 'Doe'; -- Execute the procedure to test it
```

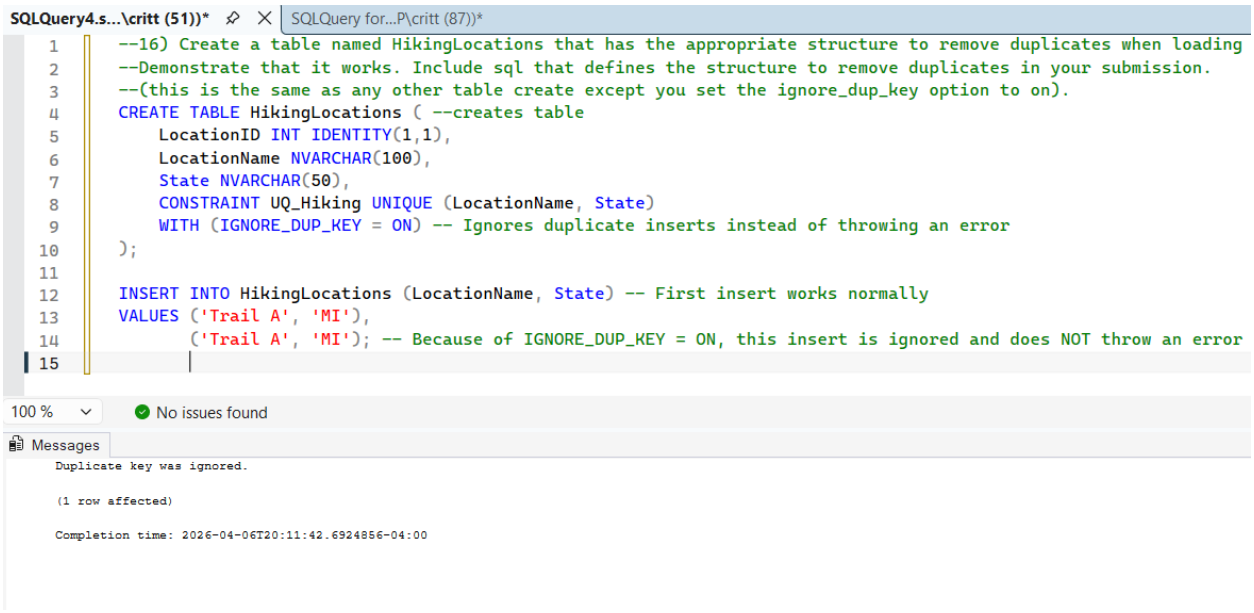
```
Commands completed successfully.

Completion time: 2026-04-06T20:06:06.1062120-04:00
```

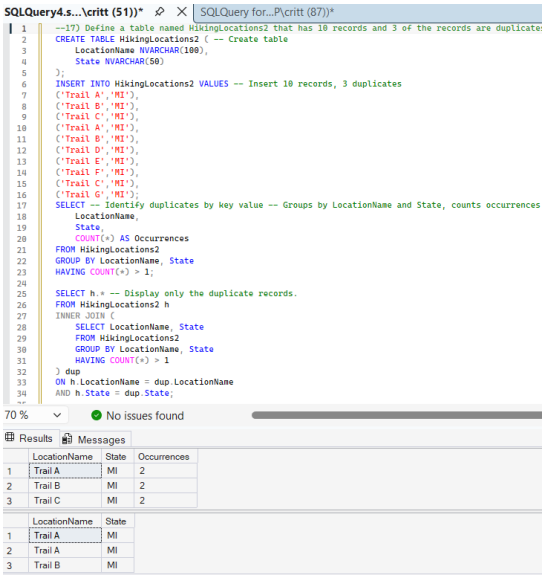
15) SQL insert statement that inserts ten rows of data into a new table named Cars that includes at least 5 columns and the data varies for each record by using variables that are incremented and/or changed within the routine for each record.



16) Create a table named HikingLocations that has the appropriate structure to remove duplicates when loading the table. Demonstrate that it works. Include sql that defines the structure to remove duplicates in your submission. (this is the same as any other table create except you set the ignore_dup_key option to on).



17) Define a table named HikingLocations2 that has 10 records and 3 of the records are duplicates. Create an SQL routine that identifies the duplicate records by key value. Create a second SQL routine that displays only the duplicate records.



18) Create an insert routine for HikingLocations2 that uses variables and performs error checking and error logging.